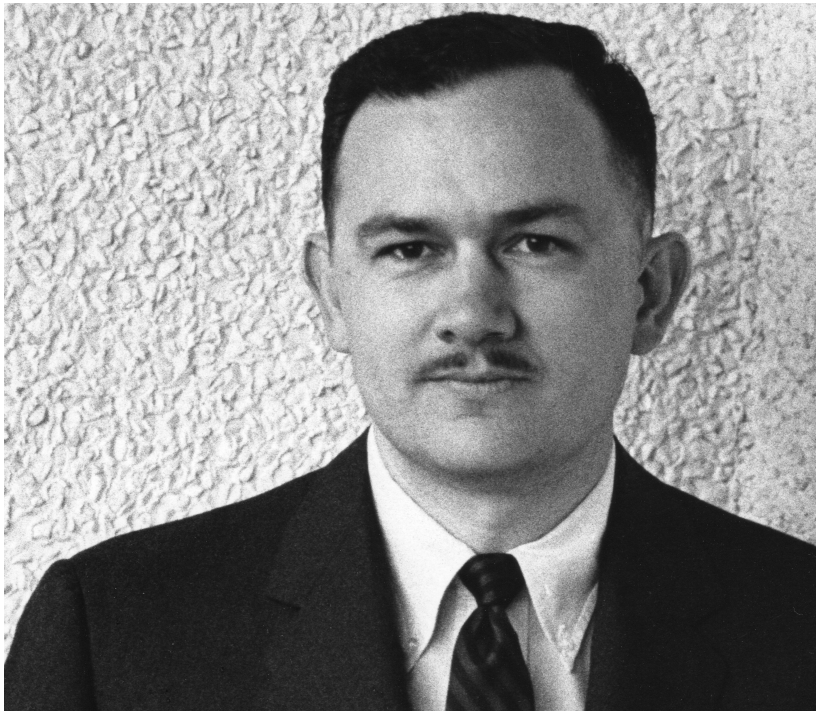
Computer Graphics in the 60s

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2026 — II Semester

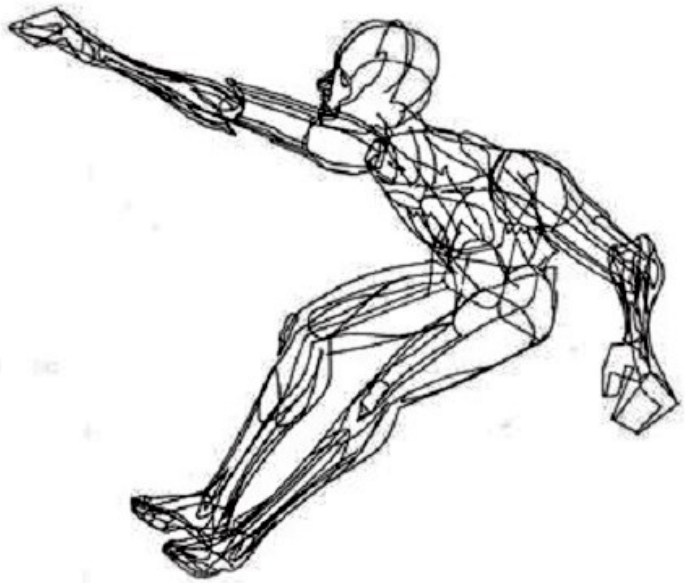
Introduction of CG Term



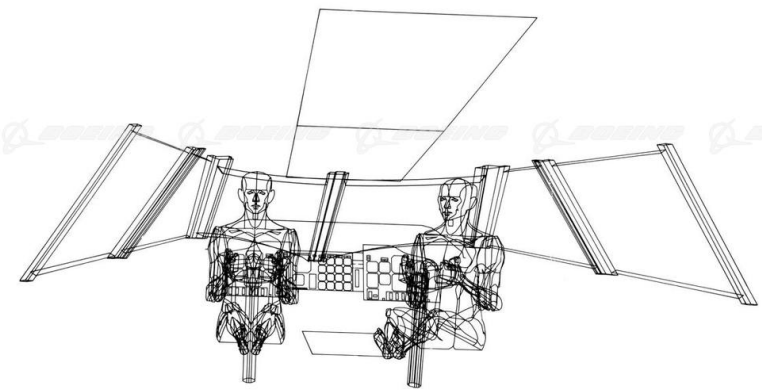
William A. Fetter coined the CG term while he was working as a graphics designer at Boeing in 1960.

First Boeing Man

He was the first person to represent the human figure as a 3D model. He created the model to study ergonomics and pilot visibility in aircraft cockpits.



First Boeing Man



Pilots visibility in cockpit

1961 - Spacewar by Steven Russell



In 1961, the MIT student Steven Russell developed Spacewar game. It was the first video game for the DEG PDP-1 minicomputer in 1962, this computer is important since it was the first commercial computer centered in direct user interaction. Thanks to the PDP-1, the hacker culture in the MIT rose.

1963 - Sketchpad by Ivan Sutherland



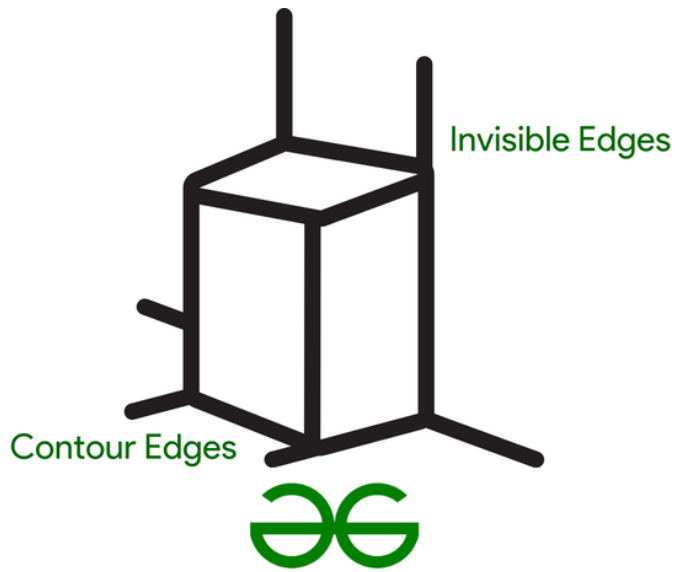
In 1963, Ivan Sutherland developed the Sketchpad for his doctoral thesis at MIT. This is important, since it was the first time that users could use a light-pen to draw shapes on a vector-based monitor. Sutherland also introduced the concept of pop-up menus.

1963 - The Mouse by Doug Engelbart



In 1963, Doug Engelbart invented the mouse at the Stanford Research Institute. The first prototype was made of a wooden shell, two wheels to track X-Y movements and a circuit-board.

1963 - The Hidden-Line Removal by Larry Roberts



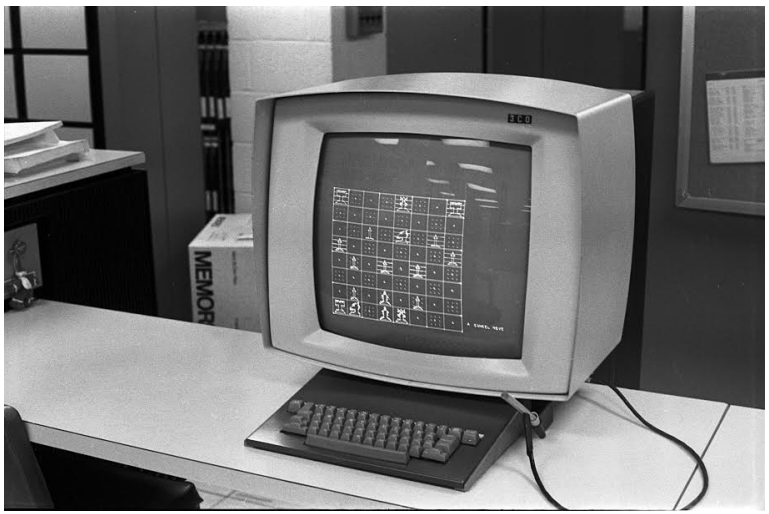
Contributing more to the 60s, Larry Roberts developed the first effective hidden-line removal algorithm. In simple words, it was an algorithm that removes the lines that should hidden in what it was supposed to be a 3D object. This algorithm helped to developed 3D objects rendering accurately.

1964 - The RAND Tablet



The RAND Corportation created the RAND tablet, also known as the Grafacon, a device intended for handwriting recognition via stylus. It was apparently the first graphic device that was: digital, low-cost, able to describe 10^6 locations on the screen, and had good linearity.

1964 - The IBM 2250



The IBM 2250 Graphics Display Unit was announced. It was made to display and interact with cloud chamber images. At the time, an entire system would cost approximately $280 000.

1965 - Jack Bresenham's paper

An algorithm is given for computer control of a digital plotter.

The algorithm may be programmed without multiplication or division instructions and is efficient with respect to speed of execution and memory utilization.

Algorithm for computer control of a digital plotter

by J. E. Bresenham

This paper describes an algorithm for computer control of a type of digital plotter that is now in common use with digital computers.[1]

The plotter under consideration is capable of executing, in response to an appropriate pulse, any one of the eight linear movements shown in Figure 1. Thus, the plotter can move linearly from a point on a mesh to any adjacent point on the mesh. A typical mesh size is 1/100th of an inch.

The data to be plotted are expressed in an (x, y) rectangular coordinate system which has been scaled with respect to the mesh; i.e., the data points lie on mesh points and consequently have integral coordinates.

It is assumed that the data include a sufficient number of appropriately selected points to produce a satisfactory representation of the curve by connecting the points with line segments, as illustrated in Figure 2. In Figure 3, the line segment connecting

Figure 1 Plotter movements

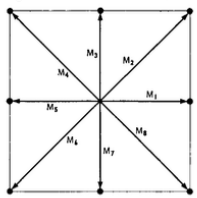
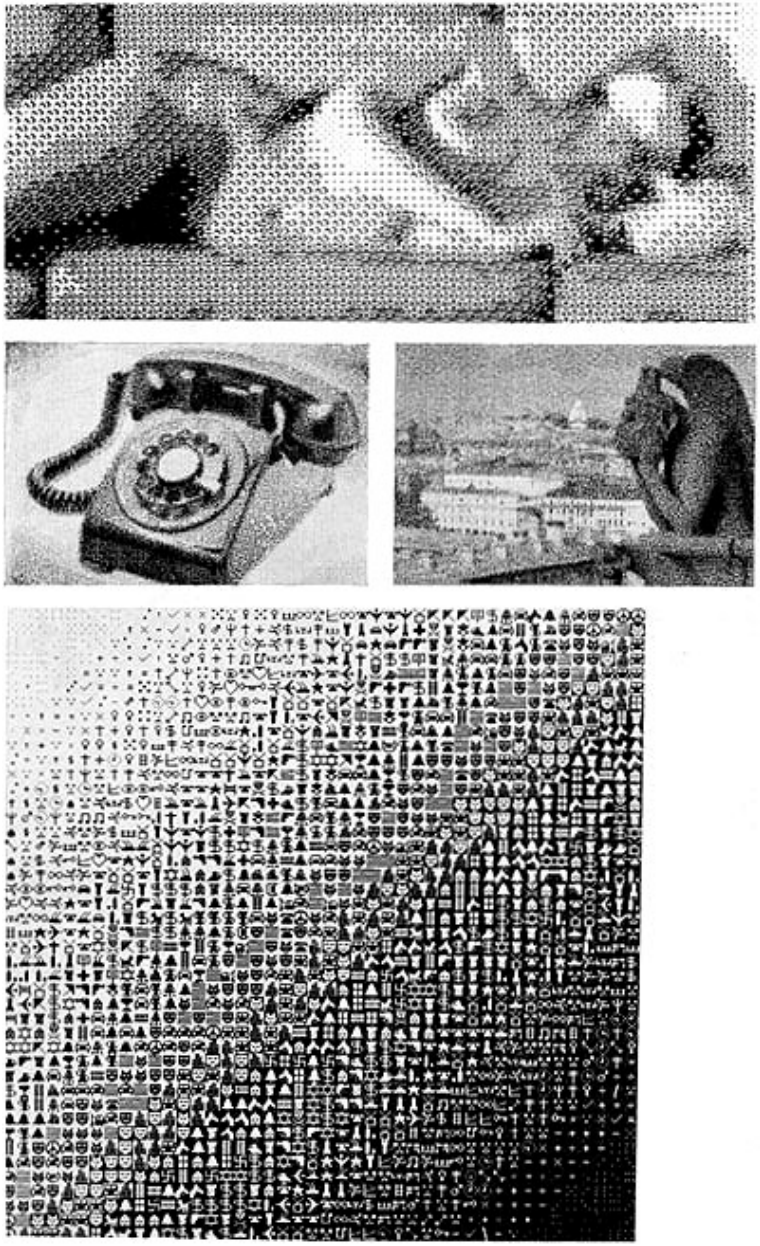


Figure 2 Curve defined by linear segments joining data points



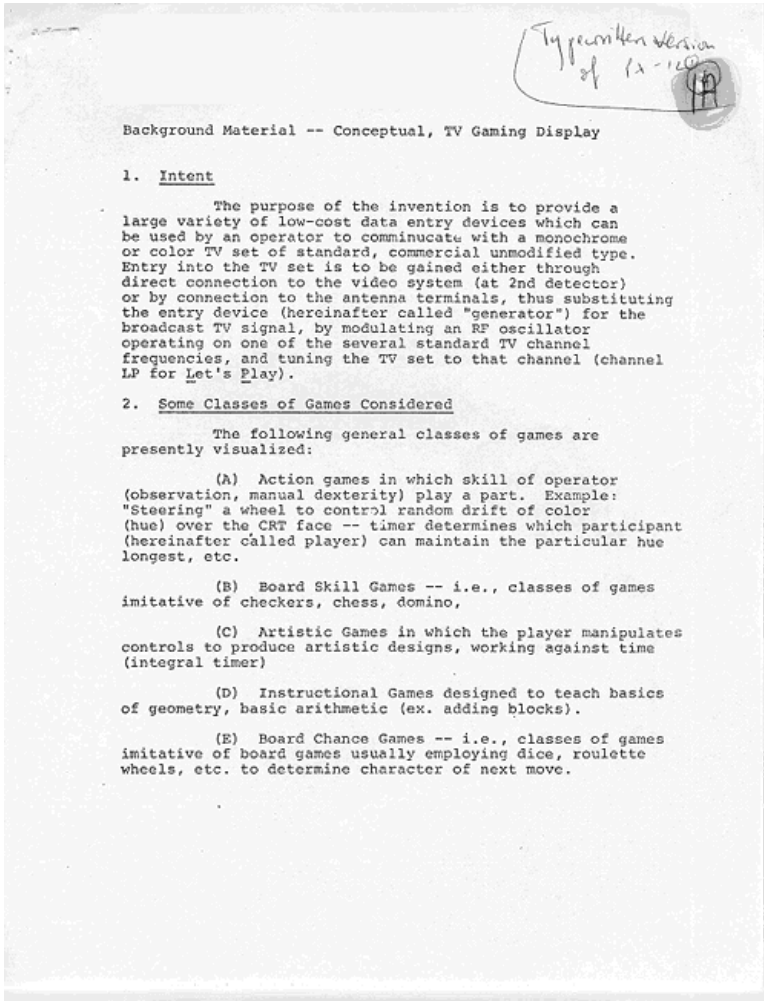
Jack Bresenham publishes a paper called "Algorithm for computer control of a digital plotter" detailing the line algorithm he developed in 1962.

1966 - ASCII art appears



Kenneth Knowlton, who worked at Bell Labs, created some of the earliest ASCII art known. He shared the images through a publication called "Studies in Perception I".

1966 - Conceptual, TV Gaming Display



Typewritten version of 1x-120

Background Material -- Conceptual, TV Gaming Display

1. Intent

The purpose of the invention is to provide a large variety of low-cost data entry devices which can be used by an operator to comminucate with a monochrome or color TV set of standard, commercial unmodified type. Entry into the TV set is to be gained either through direct connection to the video system (at 2nd detector) or by connection to the antenna terminals, thus substituting the entry device (hereinafter called "generator") for the broadcast TV signal, by modulating an RF oscillator operating on one of the several standard TV channel frequencies, and tuning the TV set to that channel (channel LP for Let's Play).

2. Some Classes of Games Considered

The following general classes of games are presently visualized:

(A) Action games in which skill of operator (observation, manual dexterity) play a part. Example: "Steering" a wheel to control random drift of color (hue) over the CRT face -- timer determines which participant (hereinafter called player) can maintain the particular hue longest, etc.

(B) Board Skill Games -- i.e., classes of games imitative of checkers, chess, domino,

(C) Artistic Games in which the player manipulates controls to produce artistic designs, working against time (integral timer)

(D) Instructional Games designed to teach basics of geometry, basic arithmetic (ex. adding blocks).

(E) Board Chance Games -- i.e., classes of games imitative of board games usually employing dice, roulette wheels, etc. to determine character of next move.

Ralph Baer, a game developer and engineer, published an article showcasing a concept of how a monochrome or color monitor could be used for developing games, as well as some possible game types as examples.baer

1966 - First HMD



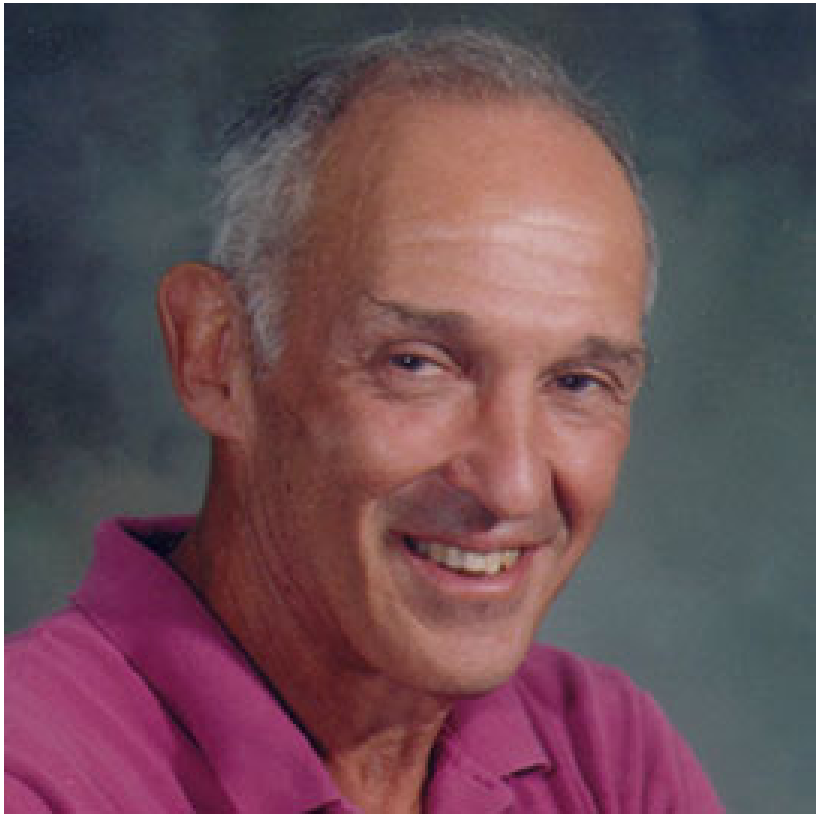
Ivan Sutherland creates the first head-mounted 3D display for augmented reality simulations. It is also called "The Sword of Damocles".

1967 - Center for Advanced Visual Studies (CAVS)



In 1967, György Kepes founded the Center for Advanced Visual Studies (CAVS) at MIT, creating a multidisciplinary community of artists, architects, engineers, and scientists. The center explored how emerging technologies could be used to create large-scale artistic works, with early fellows focusing on areas such as light art and kinetic sculpture.

1968 - Cornell's Program of Computer Graphics (PCG)



In 1968, Donald Greenberg joined Cornell University with a joint appointment in architecture and structural engineering. He established the Program of Computer Graphics, helping establish Cornell as an important center for computer graphics.

1968 - Evans and Sutherland (E&S)



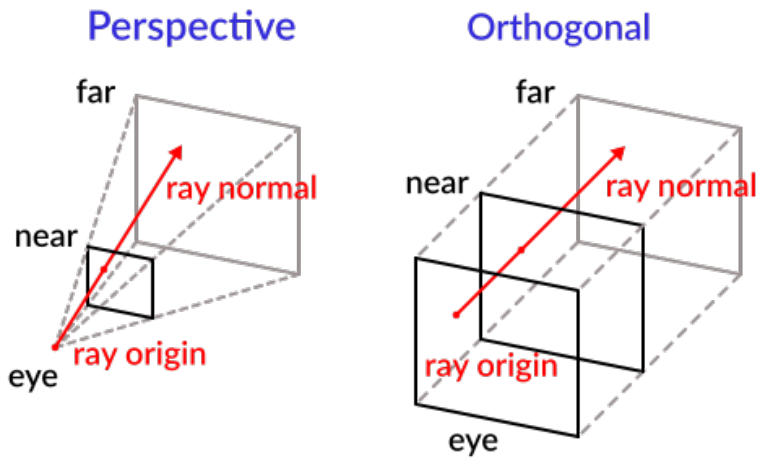
In 1968, David Evans joined the University of Utah and invited Ivan Sutherland to join him, creating an important computer graphics research group. Frustrated by the limited graphics hardware available, Evans and Sutherland founded Evans & Sutherland, one of the first companies dedicated to computer graphics.

1968 - Foundation of Intel



In 1968, Robert Noyce and Gordon Moore founded Intel (Integrated Electronics) after leaving Fairchild Semiconductor. The company was established with a strong focus on continuous innovation and research and development, becoming a major force in the development of microprocessors and hardware.

1968 - Ray-Casting



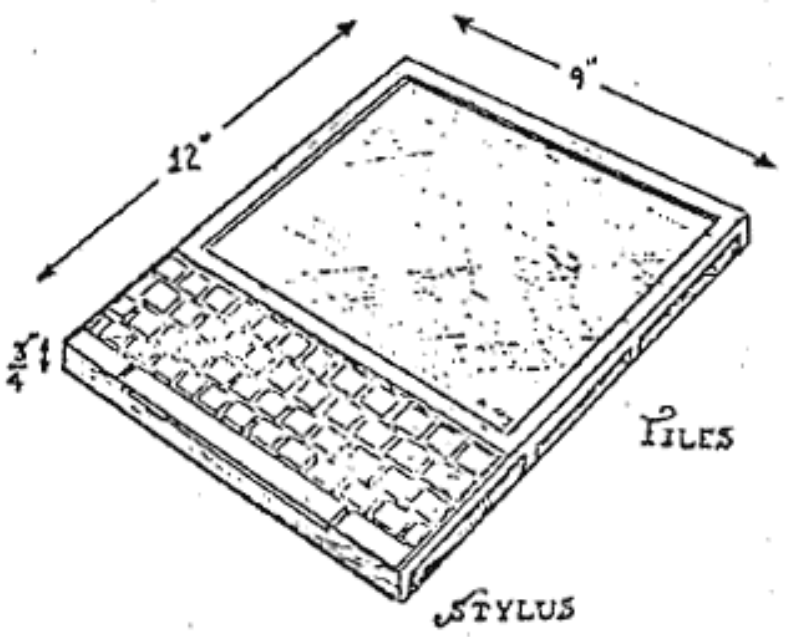
In 1968, Arthur Appel introduced the ray-casting technique to determine surface visibility in computer-generated scenes, and precursor of ray-tracing. The method casts rays from the viewer through each pixel and identifies the closest object intersected, while additional rays determine if the visible point is in shadow.

1969 - Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH)



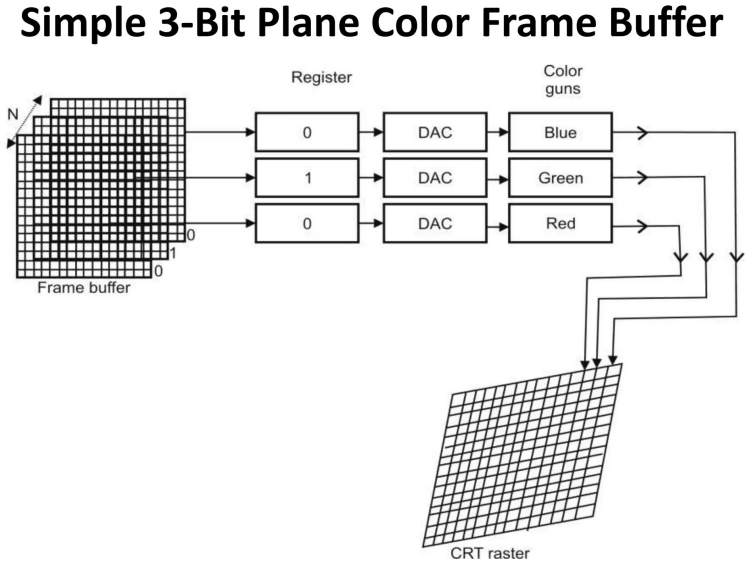
In 1969, SIGGRAPH was officially established by the ACM after Sam Matsa and Andries van Dam highlighted the growing importance of Computer Graphics. The group had originated in 1963, with its founders giving talks at various conferences and short courses.

1969 - Graphical User Interface (GUI)



In 1969, Alan Kay started envisioning a personal learning device for children, the "Dynabook". Recruited to work at Xerox PARC, he developed the idea of graphical representations of computing functions which led to the creation of operating systems with a graphical user interface, or GUI.

1969 - The First Framebuffer



In 1969, the earliest functional frame buffer was a simple 3-bit paint program developed by Joan Miller at Bell Labs. Bell engineers developed several functional raster systems under the direction of A. Michael Noll, and by 1970 they had a system capable of producing real-time half-tone images on a raster display.

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